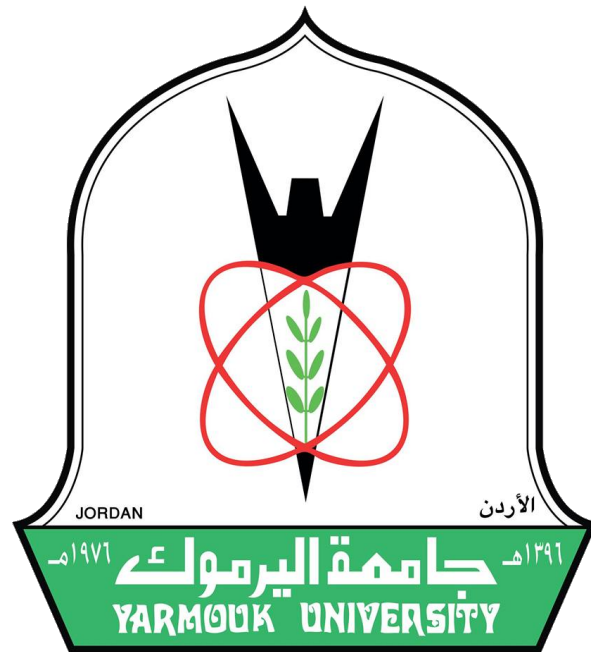


Mall Customer Segmentation



Department of Statistics

Subject: Cluster Analysis

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Cluster analysis:

is a data analysis technique that groups similar data points into “clusters,” where items within a cluster are more alike to each other than to items in other clusters, helping discover natural groupings in data without prior labels (unsupervised learning). It’s used in fields like marketing (customer segmentation) or healthcare (patient grouping) to find patterns, understand relationships, and simplify complex data sets by highlighting similarities and differences.

How it works:

- Finds natural groups: It identifies inherent structures within data, like grouping customers by buying habits or patients by symptoms.
- Measures similarity: It uses criteria like distance, density, or statistical distributions to determine how alike data points are.
- Unsupervised learning: Since it doesn’t use predefined categories, it’s great for exploring data and discovering new insights.

Key characteristics:

- High intra-cluster similarity: Objects within the same cluster are very similar.
- Low inter-cluster similarity: Objects in different clusters are dissimilar.

Common methods:

- K-Means Clustering: Divides data into a pre-specified number (k) of clusters.
- Hierarchical Clustering: Creates a tree-like hierarchy of clusters, useful for seeing relationships at different levels.

About This Data:

- This data set contains customer data that can be used for segmentation analysis. It includes demographic and behavioral information about customers, which can help in identifying distinct segments within the customer base. This can be particularly useful for targeted marketing strategies, improving customer satisfaction, and increasing sales.

we will use a cluster analysis for this data, by applying k-means methods:

- K-Means Clustering Method:
 - K-means clustering (Mac Queen, 1967) is the most commonly used unsupervised machine learning algorithm for partitioning a given data set into a set of k clusters.
 - k represents the number of clusters pre-specified by the analyst
 - It classifies such that objects within the same cluster are as similar as possible (i.e., high intra-class similarity), whereas objects from different clusters are as dissimilar as possible.

The Data View:

- our Data is very large so, we will view the first Ten rows and the last Ten rows.
- The first Ten rows :

Table 1: Mall Customer Segmentation

id	age	Gender	Income	Spending Score	Membership years	Purchase frequency	Preferred category	Last purchase amount
1	38	Female	99342	90	3	24	Groceries	113.53
2	21	Female	78852	60	2	42	Sports	41.93
3	60	Female	126573	30	2	28	Clothing	424.36
4	40	Other	47099	74	9	5	Home & Garden	991.93
5	65	Female	140621	21	3	25	Electronics	347.08
6	31	Other	57305	24	3	30	Home & Garden	86.85
7	19	Other	54319	68	5	43	Clothing	191.72
8	43	Male	108115	94	9	27	Groceries	734.56
9	53	Male	34424	29	6	7	Sports	951.71
10	55	Female	45839	55	7	2	Electronics	821.18

- The last Ten rows:

Table 2: Mall Customer Segmentation

id	age	Gender	Income	Spending Score	Membership years	Purchase frequency	Preferred category	Last purchase amount
990	31	Other	45695	36	2	31	Electronics	864.67
991	21	Other	80264	41	4	26	Sports	458.93
992	53	Other	83578	97	5	50	Sports	917.08
993	22	Female	125479	57	8	27	Home & Garden	139.75
994	29	Female	64799	42	5	19	Electronics	967.26
995	60	Male	71923	25	2	40	Clothing	986.97
996	57	Male	112170	57	6	1	Clothing	313.64
997	23	Other	65337	76	10	23	Groceries	632.83
998	23	Male	113097	40	5	42	Sports	75.09
999	22	Female	113695	63	7	44	Electronics	505.16
1000	36	Female	90420	7	2	31	Groceries	669.26

Variable type & Summary:

- Dimension for Data:

Table 3: Dimensions of the Dataset

Dimension	Value
Number of Rows	1000
Number of Columns	9

The table shows that the number of data (rows) = 1000, and the number of variables (columns) = 9 We excluded the first column (Id) because it only represents the person*.

- Variable Type:

Table 4: Structure of the Dataset

Variable	Type	Example
id	integer	1, 2, 3, 4
age	integer	38, 21, 60, 40
gender	character	Female, Female, Female, Other
income	integer	99342, 78852, 126573, 47099
spending_score	integer	90, 60, 30, 74
membership_years	integer	3, 2, 2, 9
purchase_frequency	integer	24, 42, 28, 5
preferred_category	character	Groceries, Sports, Clothing, Home & Garden
last_purchase_amount	numeric	113.53, 41.93, 424.36, 991.93

- Summary of Quantitative Variable:

Table 5: Summary of Quantitative Variable

	Age	Income	Spending Score	Membership Years	Purchase frequency	last Purchase Amount
Min.	18.000	30004.00	1.000	1.000	1.000	10.4000
1st Qu.	30.000	57911.75	26.000	3.000	15.000	218.7625
Median	45.000	87845.50	50.000	5.000	27.000	491.5950
Mean	43.783	88500.80	50.685	5.469	26.596	492.3487
3rd Qu.	57.000	116110.25	76.000	8.000	39.000	747.1700
Max.	69.000	149973.00	100.000	10.000	50.000	999.7400

- Summary For Qualitative Variable:

Table 6: Summary of Gender

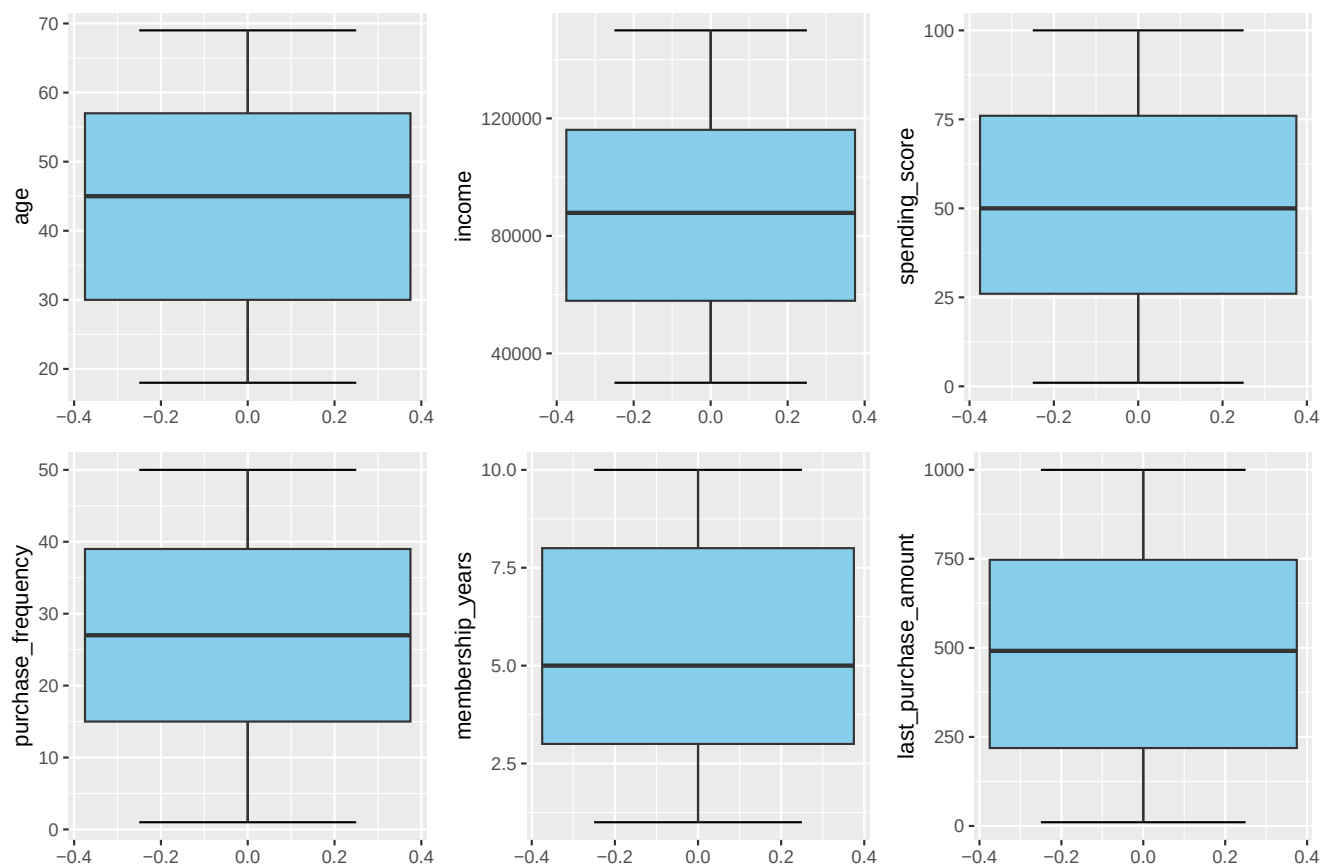
Gender	Frequency	Percentage
Female	316	31.6
Male	357	35.7
Other	327	32.7

Table 7: Summary of Preferred Category

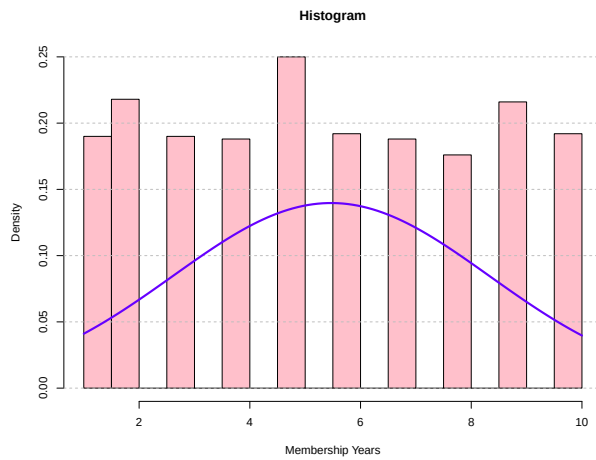
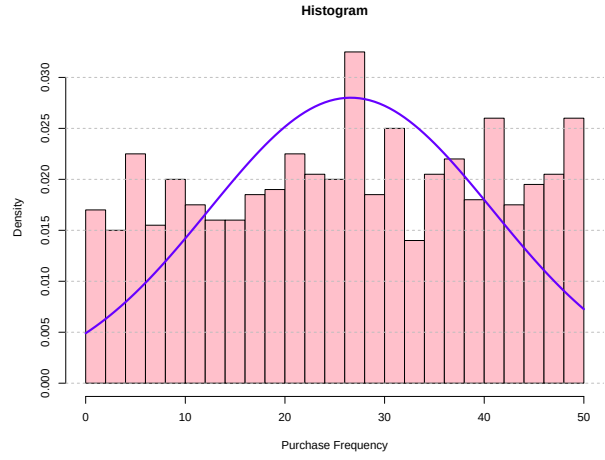
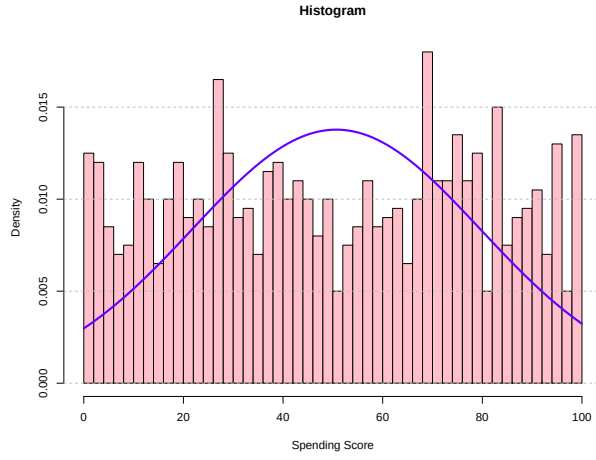
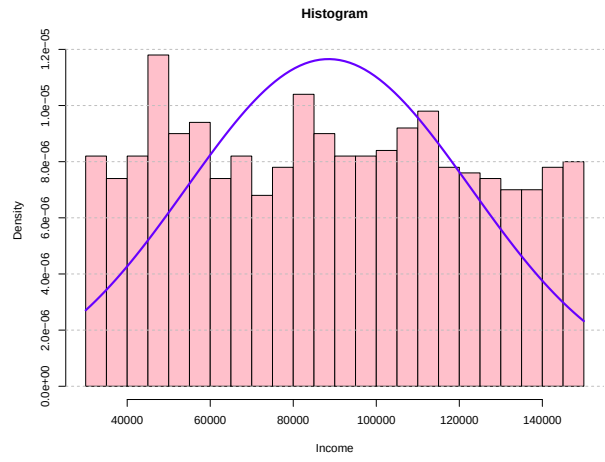
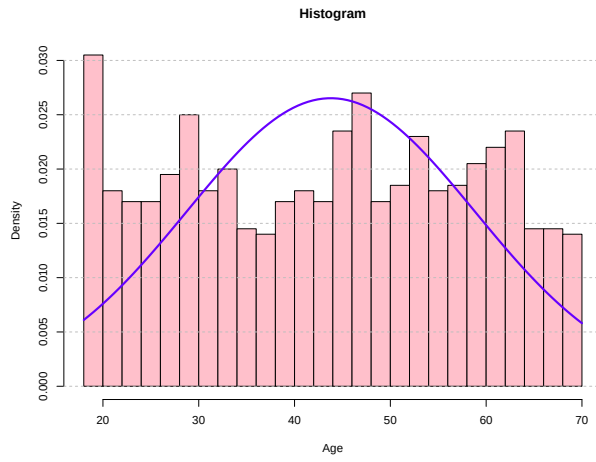
preferred_category	Frequency	Percentage
Clothing	170	17.0
Electronics	215	21.5
Groceries	199	19.9
Home & Garden	206	20.6
Sports	210	21.0

Plot for the Quntitative Variable:

- Boxplots:

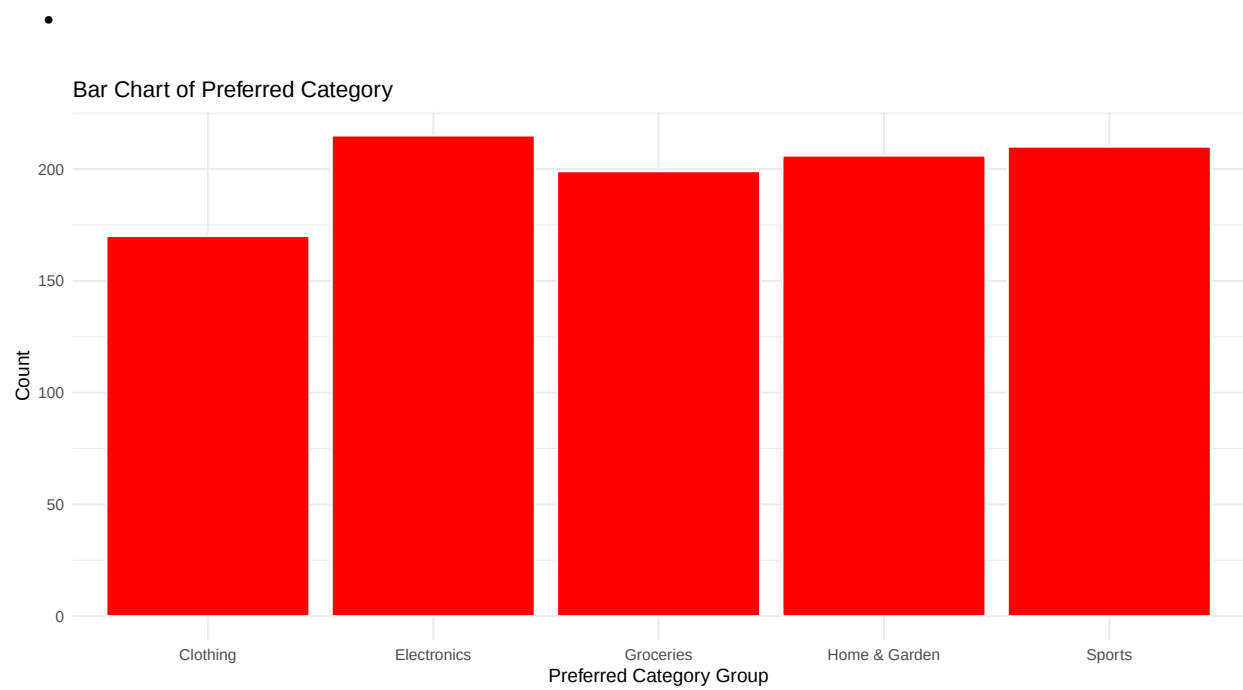
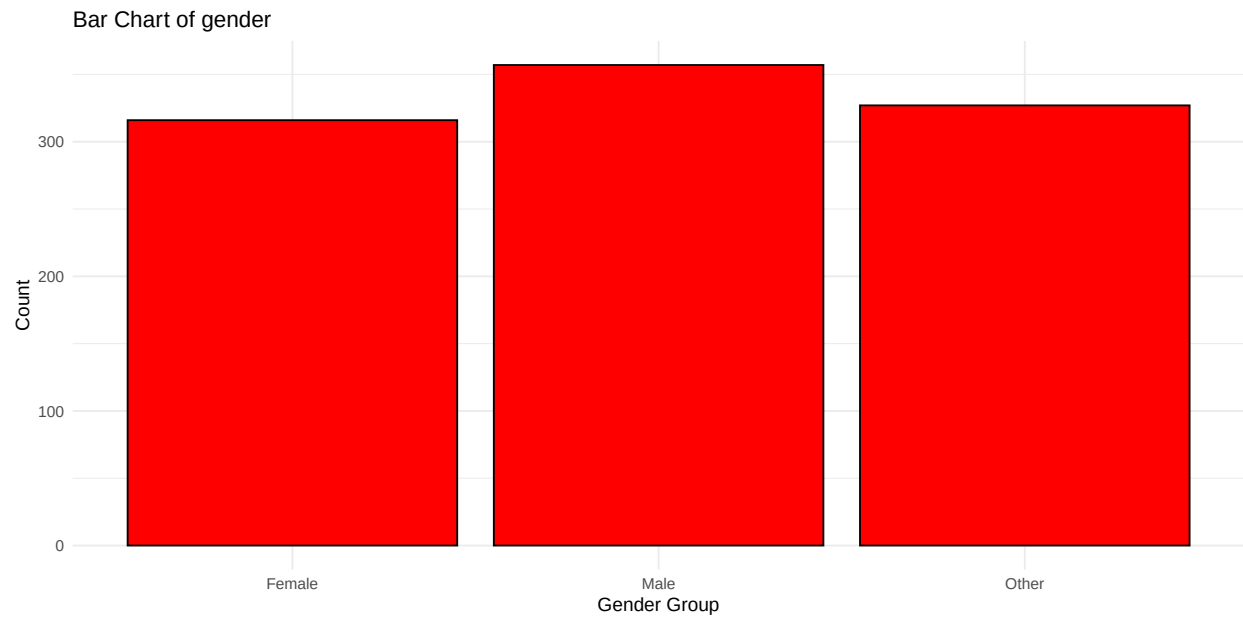


- Histogram plots:



Plot for Qualitative Variable:

- Barchart:



Applying K-Means Clustering Method:

- First, The Algorithm only works on quantitative variables, so we will only select quantitative variables in the data.
- After only select quantitative variables in the data, our Data is very large so, we will view the first Ten rows and the last Ten rows.

Table 8: Mall Customer Segmentation

id	age	Income	Spending Score	Membership years	Purchase frequency	Last purchase amount
1	38	99342	90	3	24	113.53
2	21	78852	60	2	42	41.93
3	60	126573	30	2	28	424.36
4	40	47099	74	9	5	991.93
5	65	140621	21	3	25	347.08
6	31	57305	24	3	30	86.85
7	19	54319	68	5	43	191.72
8	43	108115	94	9	27	734.56
9	53	34424	29	6	7	951.71
10	55	45839	55	7	2	821.18

- The last Ten rows:

Table 9: Mall Customer Segmentation

id	age	Income	Spending Score	Membership years	Purchase frequency	Last purchase amount
990	31	45695	36	2	31	864.67
991	21	80264	41	4	26	458.93
992	53	83578	97	5	50	917.08
993	22	125479	57	8	27	139.75
994	29	64799	42	5	19	967.26
995	60	71923	25	2	40	986.97
996	57	112170	57	6	1	313.64
997	23	65337	76	10	23	632.83
998	23	113097	40	5	42	75.09
999	22	113695	63	7	44	505.16
1000	36	90420	7	2	31	669.26

- Dimension for Data After selecting Quantitative variable :

Table 10: Dimensions of the Dataset

Dimension	Value
Number of Rows	1000
Number of Columns	6

Scale The Data:

- Now scaling The Data to ensure that all variables are measured in the same unit, resulting in a fair division, and to prevent any single variable from having a greater impact on the division than others.
- First Ten rows After scaling data :

Table 11: Scale Mall Customer Segmentation Data

age	Income	Spending Score	Membership years	Purchase frequency	Last purchase amount
-0.3844514	0.3167092	1.3577884	-0.8645775	-0.1822566	-1.2808995
-1.5146042	-0.2818750	0.3217041	-1.2147507	1.0814641	-1.5230006
1.0780993	1.1122215	-0.7143801	-1.2147507	0.0985702	-0.2298901
-0.2514922	-1.2094907	0.8052101	1.2364614	-1.5161840	1.6892343
1.4104972	1.5226125	-1.0252053	-0.8645775	-0.1120499	-0.4911969
-0.8498085	-0.9113379	-0.9215969	-0.8645775	0.2389836	-1.3711126
-1.6475634	-0.9985694	0.5979933	-0.1642312	1.1516708	-1.0165157
-0.0520535	0.5729991	1.4959329	1.2364614	0.0283635	0.8189891
0.6127423	-1.5797716	-0.7489162	0.1859419	-1.3757706	1.5532384
0.7457014	-1.2462997	0.1490234	0.5361151	-1.7268041	1.1118773

- last Ten rows After scaling data :

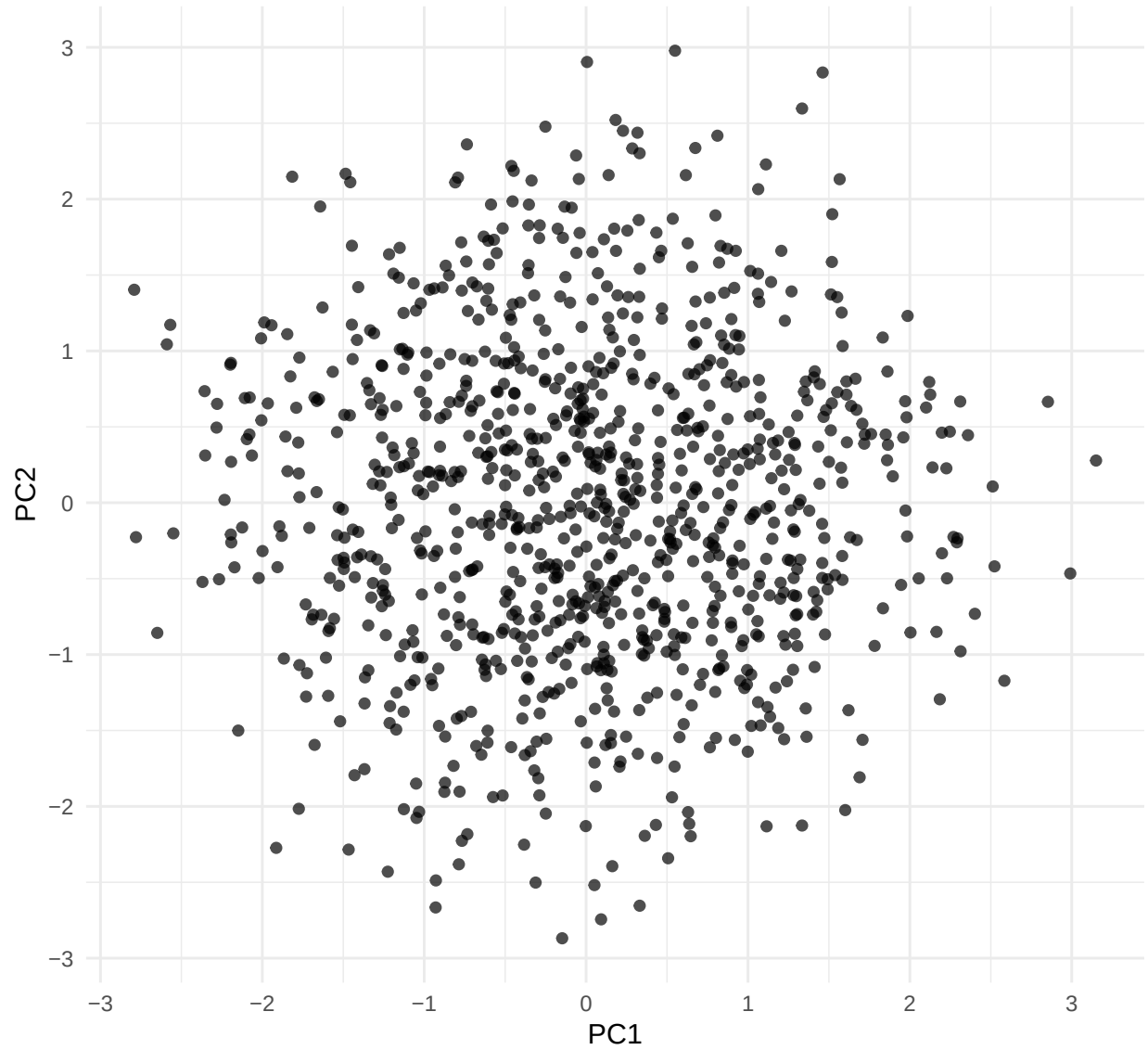
Table 12: Scale Mall Customer Segmentation Data

age	Income	Spending Score	Membership years	Purchase frequency	Last purchase amount
-0.8498085	-1.2505064	-0.5071632	-1.2147507	0.3091903	1.2589301
-1.5146042	-0.2406256	-0.3344825	-0.5144044	-0.0418432	-0.1129985
0.6127423	-0.1438121	1.5995413	-0.1642312	1.6431178	1.4361440
-1.4481247	1.0802620	0.2180957	0.8862882	0.0283635	-1.1922418
-0.9827676	-0.6924121	-0.2999464	-0.1642312	-0.5332901	1.6058176
1.0780993	-0.4842953	-0.8870608	-1.2147507	0.9410507	1.6724630
0.8786606	0.6914597	0.2180957	0.1859419	-1.7970109	-0.6042676
-1.3816451	-0.6766952	0.8742824	1.5866346	-0.2524633	0.4750095
-1.3816451	0.7185406	-0.3690187	-0.1642312	1.0814641	-1.4108767
-1.4481247	0.7360103	0.4253126	0.5361151	1.2218775	0.0433189
-0.5174106	0.0560665	-1.5087113	-1.2147507	0.3091903	0.5981903

Plot the scale data:

Data Visualization Before Clustering (PCA)

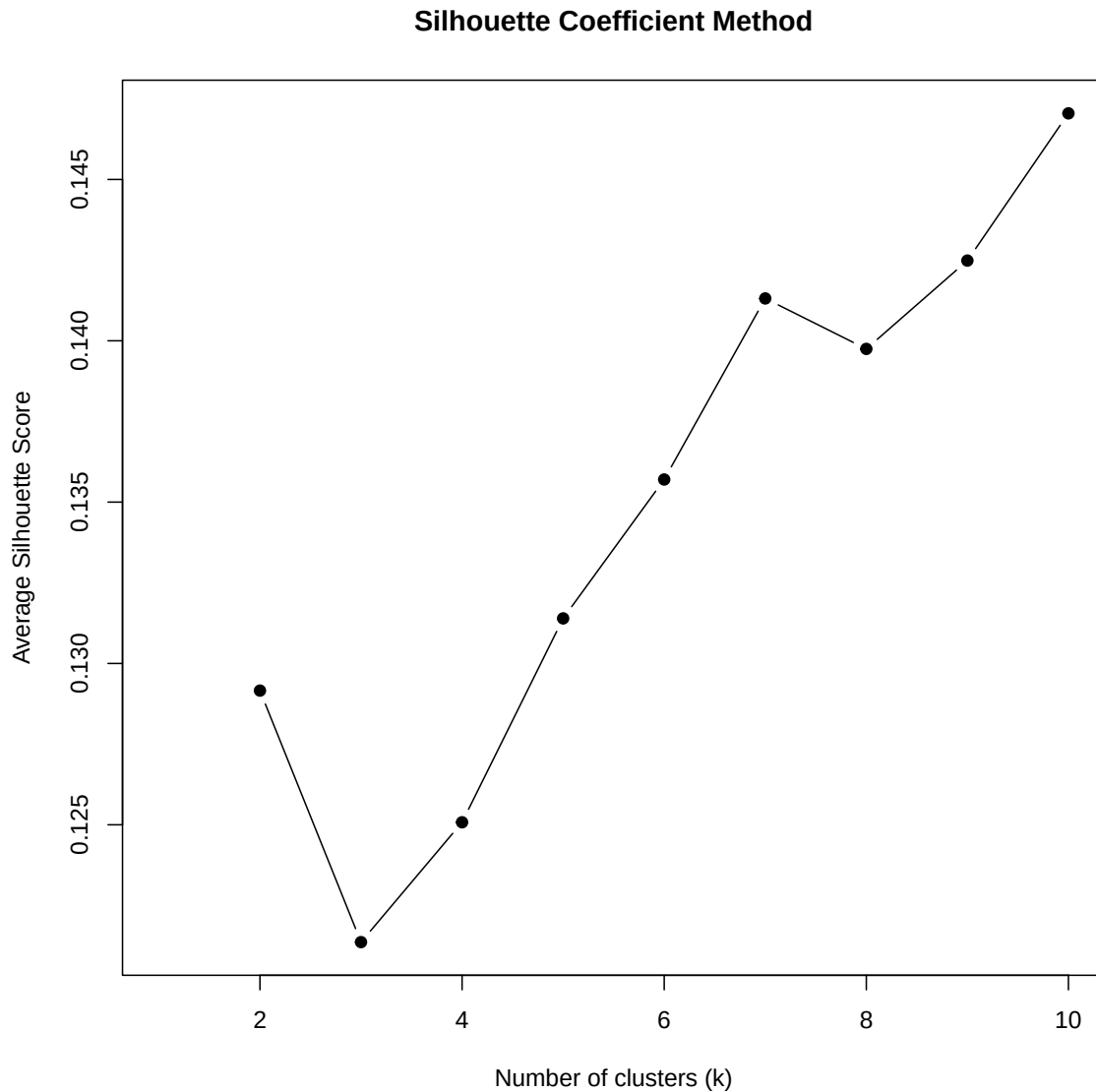
Before applying K-means



Determining the appropriate number of clusters:

There are Two method for determining the appropriate number of clusters:

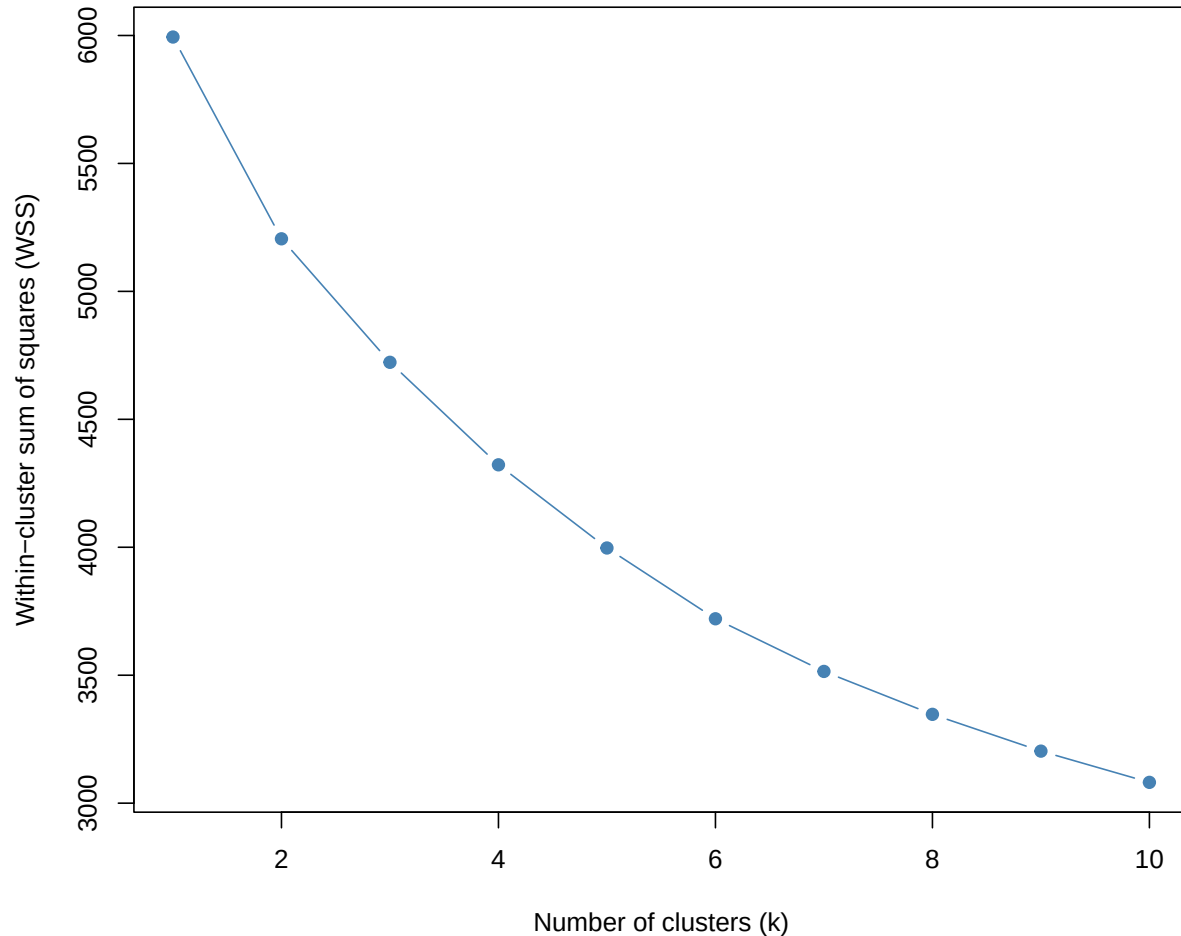
- **Silhouette Coefficient Method:** The Silhouette Coefficient Method is an internal validation technique used to evaluate clustering quality and select the optimal number of clusters. It measures how well each observation fits within its assigned cluster compared to other clusters, with values ranging from -1 to 1. Higher average silhouette values indicate better cluster separation, and the number of clusters that maximizes this average is typically considered optimal.



The Silhouette Coefficient analysis shows relatively low average silhouette values across all tested numbers of clusters, indicating weak separation among clusters. Although the highest silhouette score is observed at $k = 10$, the improvement over smaller values of k is marginal. Combined with the Elbow Method and PCA visualization, which suggest limited natural clustering structure, a five-cluster solution was selected to balance interpret ability and model simplicity.

- **Elbow Method:** The Elbow Method is a heuristic technique used to determine the optimal number of clusters in K-means clustering. It is based on calculating the within-cluster sum of squares (WSS) for different numbers of clusters and plotting these values against the corresponding number of clusters. The optimal number of clusters is identified at the point where the decrease in WSS begins to slow down significantly, forming an “elbow” in the plot. This point represents a balance between minimizing within-cluster variance and avoiding unnecessary model complexity.

Elbow Method for K-means



The Elbow Method was applied to determine the optimal number of clusters by examining the within cluster sum of squares (WSS) for different values of k . The results show a sharp decrease in WSS up to $k = 5$, followed by a more gradual reduction for larger values of k . This indicates that $k = 5$ represents the elbow point, where further increasing the number of clusters yields diminishing returns. Therefore, five clusters were selected as the optimal solution.

so, The optimal number of clusters was determined using a combination of the Elbow Method, Silhouette analysis, and PCA visualization. The Elbow Method indicated that five clusters represent the optimal balance between model complexity and within-cluster variance reduction. Although the Silhouette analysis showed relatively low values across different numbers of clusters, suggesting weak natural separation, five clusters were selected to maintain interpret ability and consistency with the Elbow criterion.

Now Applying the Algorithm:

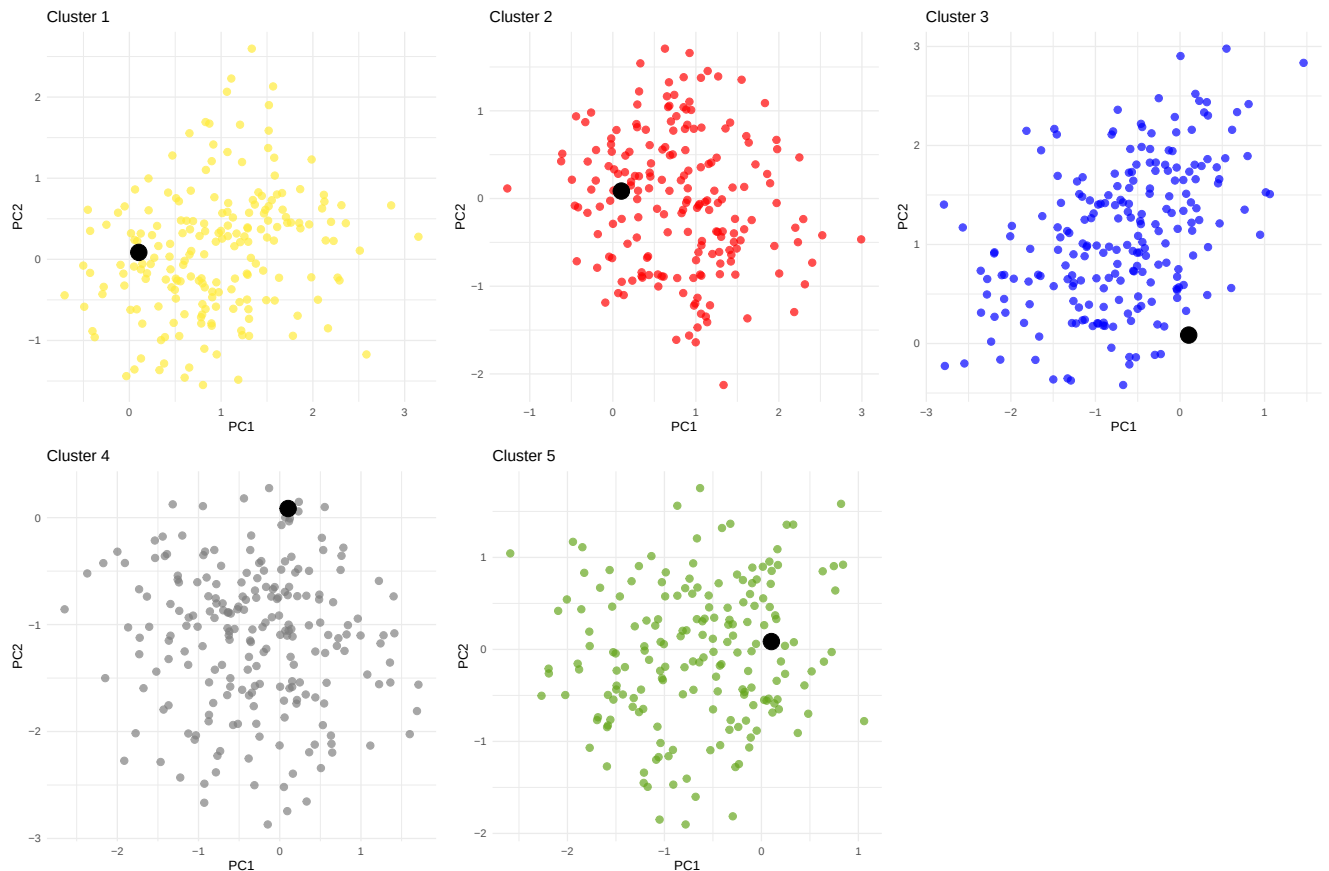
Table 13: K-means Summary Results

Metric	Value
Number of clusters (k)	5
Total within-cluster SS (tot.withinss)	3996.766
Between-cluster SS (betweeness)	1997.234
Total SS (totss)	5994
Explained variance (Between/Total)	33.32%

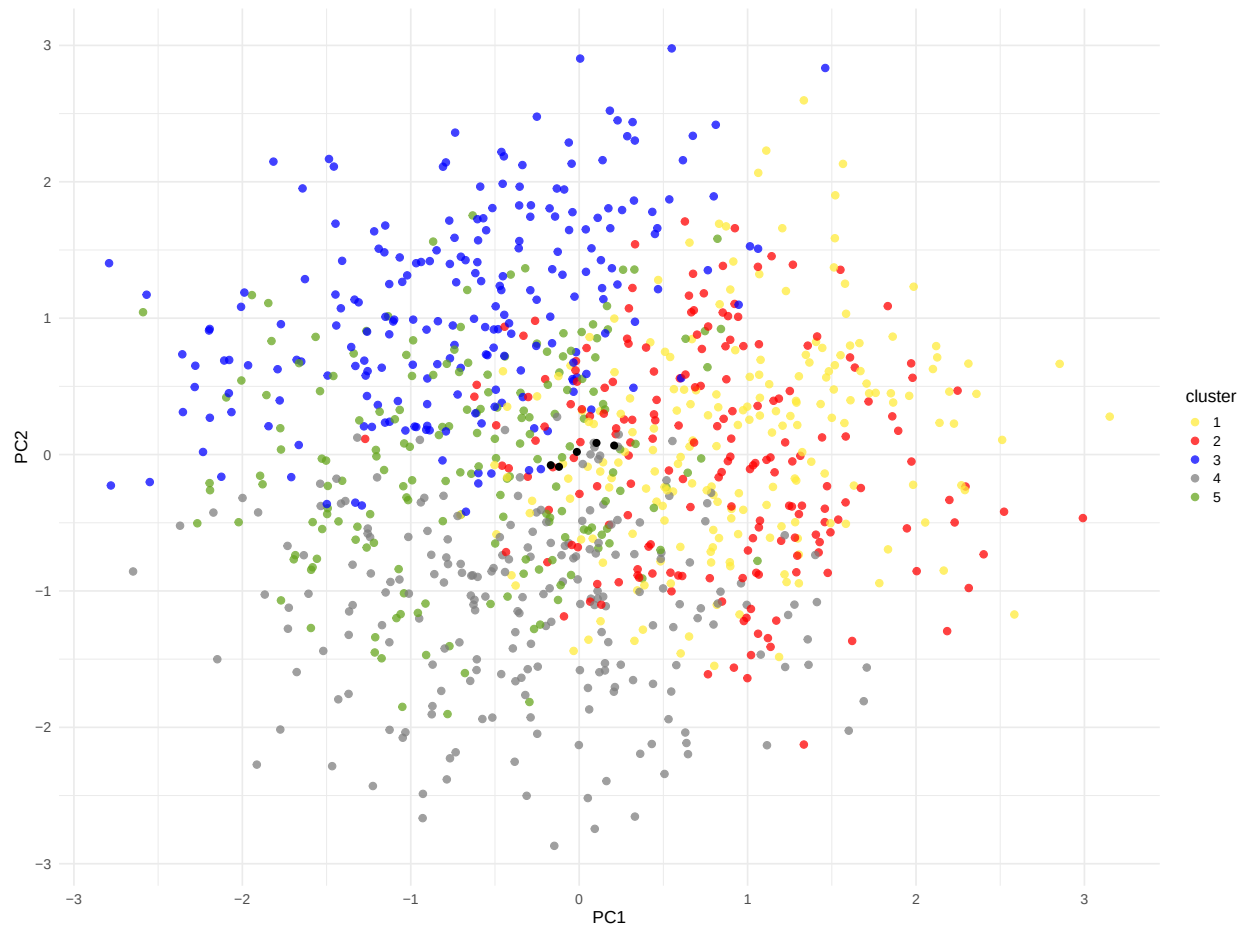
Table 14: Cluster Sizes (Number of Observations per Cluster)

Cluster	Count
1	196
2	189
3	217
4	217
5	181

Plot each cluster separately:



Plot Final Results For Applting K_means Method:



- Although K-means successfully partitioned the data into five clusters, the PCA visualization and low silhouette scores indicate substantial overlap among clusters. This suggests that the underlying data structure does not exhibit strong natural clustering, and the results should be interpreted with caution.